

“---modelName-----” : A Geo-Fenced Public Infrastructure Awareness System

Problem statement

In many cities, large-scale public infrastructure projects such as hospitals, bridges, educational institutions, and transportation facilities are developed to improve the quality of life for citizens. However, despite their significance, **most people remain unaware of the purpose, investment, and impact of these projects happening around them.**

Information related to these developments is often limited to **official reports, government portals, or news announcements**, which are not easily accessible or visible to the general public in their daily lives. As a result, citizens frequently pass by newly completed infrastructure without understanding **what was built, why it was built, and how it benefits the community.**

This gap in communication leads to **low public awareness, reduced transparency in public spending, and limited civic engagement.**

Therefore, there is a need for a **location-aware digital system that can provide citizens with clear, contextual information about nearby public development projects in real time**, enabling them to better understand the infrastructure around them and its impact on society.

Proposed Approach / Solution

Our solution introduces a **Hyper-Local Targeting Engine** that uses **geo-fencing and location-based technology** to provide citizens with real-time information about nearby public infrastructure projects. The system creates virtual geographic boundaries around important development sites such as government hospitals, bridges, colleges, and other infrastructure projects. When a user enters one of these zones, the platform automatically triggers a notification or in-app message explaining the project in simple and transparent terms.

The notification provides key details such as the **purpose of the project, investment amount, completion timeline, and the social or civic impact** it creates. Instead of requiring citizens to search government portals or reports, the information is delivered

contextually at the exact location where the development exists, making it more relevant and easier to understand.

From a technical perspective, the platform combines **GPS-based location detection, geo-spatial mapping, and a backend system that stores infrastructure data**. A mobile or web interface then delivers concise and easy-to-understand insights to users when they enter the geo-fenced area. This ensures that information about public development becomes **visible, timely, and accessible**.

What makes this idea innovative is its ability to transform **physical infrastructure into interactive information points**. Rather than static announcements or scattered reports, the system connects **real-world locations with digital civic knowledge**. This approach promotes transparency in public spending while also improving public awareness of how infrastructure investments benefit communities.

The impact of this solution is significant. It helps citizens **understand the value of public projects in their daily environment**, strengthens trust through transparent communication, and encourages greater civic engagement. At the same time, the system is highly practical because it relies on **existing technologies such as GPS, mobile notifications, and mapping APIs**, making it scalable and feasible to implement across cities and regions.

By turning everyday locations into sources of meaningful information, this platform bridges the gap between **public development and public awareness**, ensuring that citizens stay informed about the progress and impact of infrastructure around them.

Model Architecture



1. Mobile Application (User Interface)

This is the user-facing layer of the system. The mobile app collects the user's GPS location and displays notifications or information cards about nearby public infrastructure projects in a simple and accessible format.

2. Location Detection Module

This component continuously captures the user's geographic coordinates using the device's GPS services. The collected latitude and longitude data are sent to the backend system to determine whether the user is near any registered development site.

3. Geo-Fencing Engine

The geo-fencing engine creates virtual geographic boundaries around infrastructure projects. It compares the user's location with the stored coordinates and radius of each project to determine if the user has entered a defined area.

4. Infrastructure Project Database

This database stores structured information about public development projects, including location coordinates, project description, budget, completion details, and civic impact. It acts as the central data source for the system.

5. Context Processing Engine

This module converts raw project data into concise, user-friendly explanations. It ensures that the information delivered to users is clear, relevant, and easy to understand rather than technical or complex.

6. Notification Delivery System

Once the system detects that a user has entered a geo-fenced zone, this component sends a push notification or alert to the mobile application containing key details about the nearby project.

7. Admin Dashboard

The admin interface allows authorities or administrators to add new infrastructure projects, update project information, define geo-fence boundaries, and monitor user engagement with the system.

Technology Used

Mobile Application:

Developed using **Flutter or React Native** to build a cross-platform mobile app that collects user GPS location and displays notifications about nearby public infrastructure projects. **Google Maps SDK or Mapbox** will be used for map visualization.

Backend Framework:

The backend will be built with **FastAPI (Python)** to manage API requests, process location data, and implement the geo-fencing logic efficiently.

Database & Geospatial Processing:

PostgreSQL with PostGIS will be used to store infrastructure project data and perform geospatial queries to detect nearby locations accurately.

Geo-Fencing & Location Services:

The system will use **GPS-based location tracking** and distance calculation algorithms such as the **Haversine formula** to determine whether a user has entered a predefined geographic boundary.

Notification System:

Firebase Cloud Messaging (FCM) will be integrated to deliver real-time push notifications when users enter geo-fenced zones around infrastructure sites.

AI / Context Processing:

Natural Language Processing (NLP) will be used to convert technical project details into clear, easy-to-understand summaries for users, improving accessibility and engagement.

Development & Deployment Tools:

Tools such as **Git/GitHub for version control** and **Docker or cloud platforms (AWS/GCP/Azure)** for deployment will support scalable development and system management.

FEATURE / USP

Hyper-Local Geo-Fencing:

Uses precise geo-fencing to detect when users enter the vicinity of a public infrastructure project and trigger relevant notifications.

Context-Aware Notifications:

Delivers clear, location-based information explaining the purpose, investment, and impact of nearby public projects.

Real-Time Civic Awareness:

Provides instant insights about development projects around users without requiring them to search government sources.

AI-Powered Information Simplification:

Uses Natural Language Processing (NLP) to convert complex project data into simple, easy-to-understand summaries.

Interactive Map Interface:

Allows users to view nearby infrastructure projects and explore development activity within their area.

Transparency and Engagement:

Improves public awareness and transparency by clearly showing how infrastructure projects benefit the community.

Ref links, github repo , api etc etc